

Name:

Chapter 1 Atoms and elements review quiz

Multiple choice

Identify the choice that best completes the statement or answers the question.

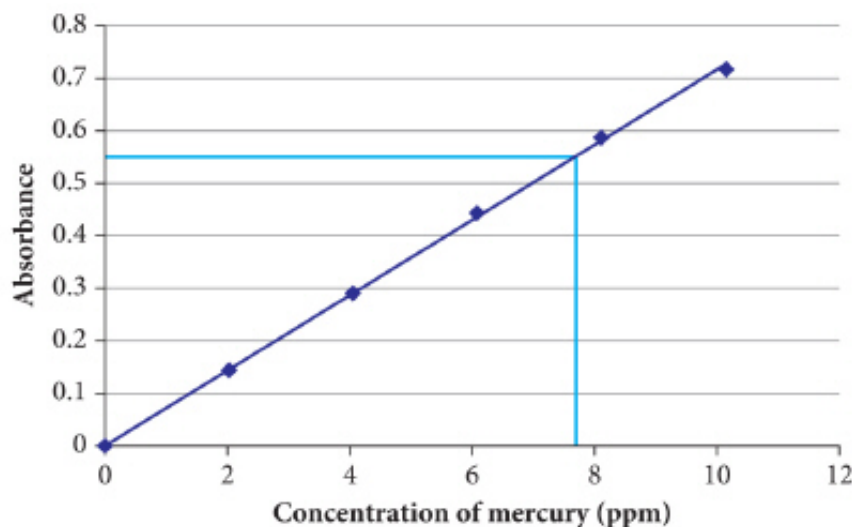
- ☐ 1 Which is the smallest subatomic particle in the atom?
- A Proton
 - B Neutron
 - C Electron
 - D Nucleus
- ☐ 2 A substance made up of only one type of atom is called:
- A a symbol.
 - B a nucleus.
 - C an electron cloud.
 - D an element.
- ☐ 3 Which of the following is the correct definition for the mass number, A , of an element?
- A The number of protons in the nucleus of an atom of an element
 - B The number of protons and neutrons in the nucleus of an atom of an element
 - C The number of neutrons in the nucleus of an atom of an element
 - D The number of electrons that orbit the nucleus of an atom of an element
- ☐ 4 Which type of force holds the atom together?
- A Electrostatic attraction
 - B Electrostatic repulsion
 - C Strong nuclear force
 - D Electromagnetic attraction
- ☐ 5 Which of the following shows the correct number of protons, neutrons and electrons, respectively, for the element sodium?
- A 11, 11, 12
 - B 11, 23, 11
 - C 11, 12, 12
 - D 11, 12, 11
- ☐ 6 An unknown element X has 6 protons and 6 neutrons. Which of the following is an isotope of element X?
- A Carbon-12
 - B Carbon-14
 - C Boron-11
 - D Nitrogen
- ☐ 7 The elements potassium, calcium, iron and bromine are all:
- A metals.
 - B in the same group.
 - C in the same period.
 - D transition elements.
- ☐ 8 Which element on the periodic table has an electron configuration 2,8,2?
- A Magnesium
 - B Sodium

- C Neon
- D Aluminium

- 9 Which of the following is the correct electronic configuration, using subshell notation, for silicon-14?
- A $1s^2 2s^2 2p^6 3s^2 3p^2$
 - B $1s^2 2s^2 2p^8 3s^2$
 - C $1s^2 2s^2 3s^2 2p^6 3p^2$
 - D 2,8,4
- 10 The electron configuration 2,7,3 is incorrect. If the correct element is in group 3 of the periodic table, which of the following is the correct electron configuration?
- A 2,8
 - B 2,8,1
 - C 2,8,2
 - D 2,8,3
- 11 Which of the following correctly compares the atomic radius and electronegativity of Cs and Ba?
- A Ba has a smaller atomic radius than Cs, and Cs is more electronegative than Ba.
 - B Cs has a smaller atomic radius and is more electronegative than Ba.
 - C Ba has a larger atomic radius and is more electronegative than Cs.
 - D Cs has a larger atomic radius than Ba, and Ba is more electronegative than Cs.
- 12 Which of the following elements are all non-metals?
- A Carbon, aluminium, oxygen
 - B Sodium, beryllium, calcium
 - C Phosphorus, iodine, neon
 - D Boron, carbon, nitrogen
- 13 The most electronegative element on the periodic table is:
- A fluorine.
 - B francium.
 - C lithium.
 - D asatine.
- 14 Which of the following statements about the ionisation energy of Al is correct?
- A Al has a lower ionisation energy than Ga.
 - B Al has a higher ionisation energy than B.
 - C Al has a lower ionisation energy than Na.
 - D Al has a higher ionisation energy than Na.
- 15 Which of the following is correct?
- A A lower energy level is closer to the nucleus.
 - B Electrons can only move from a lower energy level to a higher energy level.
 - C Electrons can only move from a higher energy level to a lower energy level.
 - D When an atom absorbs heat energy it is released again in the form of kinetic energy.
- 16 A student was conducting flame tests on six different unknown samples of nitrate and chloride salts. Two of the samples produced a pale green flame. What can the student correctly infer from these results, assuming the experiment was carried out correctly?
- A Two of the six samples are the same.
 - B Two of the six samples contain the same cation.
 - C The wire loop was not correctly cleaned between tests.
 - D The two salts emit the same set of wavelengths, and thus appear the same colour.
- 17 Which of the following is correct for atomic absorption spectroscopy (AAS)?

- ☐ A AAS is a quantitative technique only.
- ☐ B The equipment used in AAS directly measures the concentration of an element in a sample.
- ☐ C The sample being tested needs to be vaporised so that gaseous atoms are present.
- ☐ D None of the above are true.

☐ 18 Refer to the graph below.



The concentration of a mercury sample that produced an absorbance reading of 0.4 is closest to:

- ☐ A 0.3 ppm.
- ☐ B 5.0 ppm.
- ☐ C 5.5 ppm.
- ☐ D 6.0 ppm.

☐ 19 Which method of analysing elements is not based on promoting electrons to a higher energy level?

- ☐ A Flame test
- ☐ B Mass spectrometry
- ☐ C Emission spectroscopy
- ☐ D Atomic absorption spectroscopy

☐ 20 Which piece of apparatus in mass spectrometry is responsible for deflecting the charged ions?

- ☐ A The ion repeller
- ☐ B The electron gun
- ☐ C The electromagnet
- ☐ D The slit

